

Bluetooth Controlled RC Car Using Arduino Nano

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Abstract:

Bluetooth-controlled robotic vehicles have gained significant attention due to their simplicity, low cost, and wide range of applications. This paper presents the design and implementation of a Bluetooth Controlled RC Car using an Arduino Nano microcontroller. The system is controlled wirelessly through an Android smartphone using a Bluetooth Controller application. The HC-05 Bluetooth module enables communication between the smartphone and the vehicle. Motor control is achieved using the L298N motor driver module, which drives BO motors connected to rubber BO wheels. The system is powered using two 3.7 V 18650 lithium-ion batteries connected in series. A custom wooden base is used as the chassis, providing mechanical stability. The proposed system demonstrates an efficient, low-cost, and user-friendly approach to wireless vehicle control suitable for educational and hobbyist applications.[1]

Keywords:

Arduino Nano, Bluetooth Control, HC-05, L298N Motor Driver, RC Car, Android Application

I. Introduction

Wireless control systems have become an integral part of modern electronics and robotics. Among various wireless technologies, Bluetooth offers a reliable, low-power, and cost-effective solution for short-range communication. Bluetooth-controlled vehicles are widely used in robotics education, surveillance prototypes, and automation systems.

This project focuses on the development of a Bluetooth Controlled RC Car using an Arduino Nano as the main controller. The vehicle receives control commands from an Android smartphone via a Bluetooth connection. The simplicity of the system makes it ideal for beginners while still providing a strong foundation for understanding embedded systems, motor control, and wireless communication.

II. System Overview

The system consists of a smartphone, Bluetooth module, microcontroller, motor driver, motors, and power supply. The Android smartphone sends directional commands through a Bluetooth Controller application. These commands are received by the HC-05 Bluetooth module and forwarded to the Arduino Nano via serial communication.

Based on the received command, the Arduino Nano processes the data and generates appropriate control signals for the

L298N motor driver. The motor driver then controls the speed and direction of the BO motors, enabling the RC car to move forward, backward, left, and right.

III. Hardware Components

- Arduino Nano (ATmega328P microcontroller)
- HC-05 Bluetooth module
- L298N dual H-bridge motor driver
- BO DC gear motors with rubber wheels
- Two 3.7 V 18650 lithium-ion batteries with holder
- 300-pin breadboard
- Custom wooden chassis
- Jumper wires

IV. Hardware Implementation and Assembly Procedure

The hardware assembly of the Bluetooth controlled RC car is carried out in a systematic manner as described below.

Step 1: Identification of Components

First, unpack the robot car kit and carefully identify all the components provided in the box. This includes the Arduino Nano board, BO motors, motor driver module, wheels, battery holder, breadboard, jumper wires, and mechanical parts required for assembly.[2]



Fig. 1. Hardware components used in the Bluetooth controlled RC car

Step 2: Additional Components Required

Next, identify and collect the additional electronic components required to control the car. These include the HC-05 Bluetooth module, L298N motor driver, 18650 lithium-ion batteries with battery holder, jumper wires, and a 300-pin breadboard.

Step 3: Chassis Preparation

Now, assemble the robot car chassis. In this project, a custom wooden base is used as the chassis. If acrylic parts are used, remove the protective stickers carefully before assembly to ensure proper fitting and appearance.

Step 4: Motor Wiring and Code Wheel Attachment

Solder the connecting wires to the terminals of the BO gear motors. After soldering, attach the rubber wheels to the motor shafts securely to ensure smooth rotation.



Step Fig. 2. BO DC gear motor with rubber wheel

5: Mounting the Motors on the Chassis

Insert the fasteners into the chassis and mount the gear motors using screws. Ensure the motors are firmly fixed and aligned properly to avoid vibration during movement.



Fig. 3. Mounting of BO DC motors on the wooden chassis

Step 6: Caster Wheel Installation

Attach the spacers to the caster wheel and mount it on the front or rear side of the chassis. The caster wheel helps in maintaining balance and smooth turning of the robot car.



Fig. 4. Caster wheel used for chassis balance

Step 7: Battery Holder Installation

Mount the battery holder onto the chassis using screws or adhesive. In this project, two 3.7 V 18650 lithium-ion

batteries are used to provide sufficient voltage for motor operation.



Fig. 5. Installation of battery holder on the chassis

Step 8: Arduino and Bluetooth Module Placement

Place the Arduino Nano and HC-05 Bluetooth module on the 300-pin breadboard. Connect the Bluetooth module to the Arduino Nano using jumper wires according to the circuit diagram (TX–RX and RX–TX connections).[3]

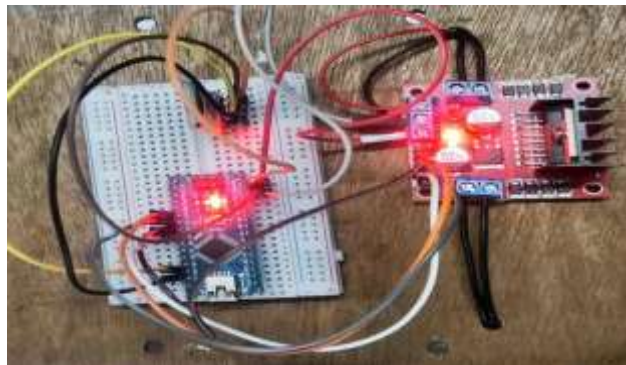


Fig. 6. Arduino Nano and HC-05 Bluetooth module placement on breadboard

Step 9: Power and Control Connections

Connect the battery holder to the power input of the L298N motor driver. Then, connect the control input pins of the motor driver to the Arduino Nano digital pins as per the circuit diagram.

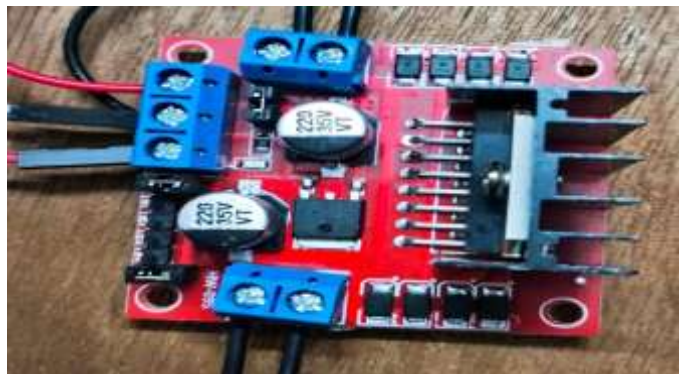


Fig. 7. L298N motor driver module connection with DC motors

Step 9: Program Uploading

Connect the Arduino Nano to the computer using a USB cable. Open the Arduino IDE and upload the Bluetooth control program.

Before uploading, disconnect the RX and TX jumper wires. After successful uploading, reconnect the RX and TX connections.



Fig. 8. Program uploading to Arduino Nano using Arduino IDE

Step 10: App Installation and Bluetooth Pairing

Remove the USB cable and insert the batteries into the holder. Install the Serial Bluetooth Terminal app from the Play Store. Open the app, navigate to the controller section, and enable Bluetooth.

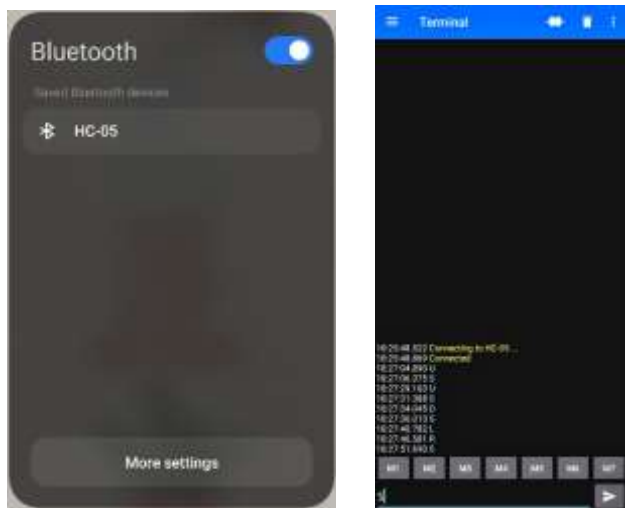


Fig. 9. Bluetooth controller application interface on Android smartphone

Step 11: Fully assembled RC car prototype

The hardware implementation of the Bluetooth controlled RC car was carried out in a systematic and well-planned manner to ensure reliable operation and ease of assembly. Initially, all the electronic and mechanical components required for the system were identified, inspected, and verified before beginning the assembly process. These components included the Arduino Nano microcontroller board, HC-05 Bluetooth module, L298N dual H-bridge motor driver, DC BO gear motors with rubber wheels, 18650 lithium-ion batteries with battery holder, breadboard, jumper wires, and a custom wooden chassis. Careful verification at this stage helped to minimize wiring errors and ensured compatibility among the components.

A custom wooden base was selected and prepared as the chassis of the RC car due to its strength, availability, and ease of modification. The positions for mounting the motors, battery holder, and control modules were marked to maintain

proper weight distribution and mechanical balance. In the first stage of assembly, the DC BO gear motors were wired and securely mounted onto the chassis using screws. Rubber wheels were attached to the motor shafts to provide adequate traction and smooth movement. A caster wheel was installed at the front side of the chassis to support balance and facilitate smooth turning during operation.



Fig. 10. Fully assembled Bluetooth controlled RC car prototype

In the next stage, the power supply unit was installed. A battery holder containing two 3.7 V 18650 lithium-ion batteries connected in series was mounted on the chassis. This configuration provided sufficient voltage and current to drive both the motors and the control circuitry. Proper insulation and secure mounting were ensured to avoid short circuits and mechanical instability during motion.

After completing the mechanical assembly, attention was given to the electronic control system. The Arduino Nano microcontroller and the HC-05 Bluetooth module were placed on a 300-pin breadboard to allow easy interconnections and modifications. Serial communication connections were established between the Bluetooth module and the Arduino Nano using jumper wires. The HC-05 module was configured to receive control commands wirelessly from an Android smartphone.

The L298N dual H-bridge motor driver module was then installed on the chassis and connected to the DC motors. Control input pins of the motor driver were interfaced with the digital output pins of the Arduino Nano, enabling bidirectional control of the motors. The power input of the motor driver was connected to the battery supply, while a common ground connection was maintained between all components to ensure stable operation.

Once all electrical connections were completed, the system was thoroughly inspected to verify correct wiring, polarity, and voltage levels. The Arduino Nano was then connected to a computer, and the control program was uploaded using the Arduino IDE. During program uploading, the RX and TX

connections of the Bluetooth module were temporarily disconnected to avoid communication conflicts. After successful uploading, the connections were restored, and the Bluetooth communication was tested using an Android Bluetooth Controller application.

Finally, all components were integrated to obtain the fully assembled Bluetooth controlled RC car prototype. Functional testing was performed to verify forward, backward, left, right, and stop operations. The system demonstrated stable wireless control, smooth motor operation, and reliable performance within the effective Bluetooth range. The completed prototype is shown in Fig. 11, representing the successful implementation of the proposed system.

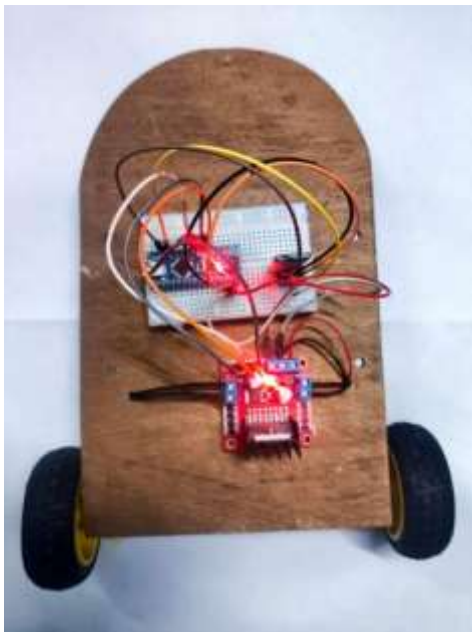


Fig. 11. Fully assembled Bluetooth controlled RC car prototype.2

V. Working Principle

After powering the system, the HC-05 Bluetooth module pairs with the Android smartphone. Control commands such as forward, backward, left, right, and stop are transmitted wirelessly. The Arduino Nano processes these commands and activates the corresponding motor driver inputs to control motor direction.

VI. Software Implementation

The Arduino Nano is programmed using the Arduino IDE. Serial communication is established at 9600 baud rate. Based on the received characters from the Bluetooth module, corresponding motor control functions are executed.

(The program code provided in the previous section is used.)

VII. Results and Discussion

The Bluetooth controlled RC car operates smoothly within a Bluetooth range of approximately 10 meters. The response

time is minimal, and the system performs reliably under normal operating conditions. The wooden chassis provides adequate mechanical strength while keeping the overall cost low.

VIII. Applications

- Educational robotics projects
- Smartphone-controlled vehicles
- Wireless automation demonstrations
- Basic surveillance robot prototypes

IX. Advantages and Limitations

A. Advantages

- Low cost and easy implementation
- Wireless control using widely available smartphones
- Compact design using Arduino Nano
- Simple circuit and programming

B. Limitations

- Limited Bluetooth range
- No obstacle detection or feedback system
- Manual control only

X. Conclusion and Future Scope

The Bluetooth controlled RC car demonstrates an effective implementation of wireless vehicle control using Arduino Nano. The project is suitable for beginners and academic demonstrations. Future improvements may include obstacle detection sensors, camera modules, and Wi-Fi-based control for extended range.

References

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